

PropertyPilot — Unit Test Results

Project: PropertyPilot (WGU D424 Capstone) **Author:** Michael Riehm **Run date:** 2026-06-01 **Tester:** Michael Riehm
Framework: Vitest 4.1.6 **Command:** npm run test -w backend

1. Headline Results

Metric	Value
Test files	30 passed , 0 failed
Individual tests	209 passed , 0 failed
Duration	~0.6 s (test phase)
Status	 All green

All 209 unit tests pass on a clean checkout with no environment setup beyond `npm install`. The suite uses mocked Prisma and JWT services, so no database or network access is required.

2. Execution Evidence

2.1 Summary view

The terminal summary printed at the end of `npm run test -w backend` :

```
PS C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone> npm run test -w backend -- --run
```

```
> @propertyilot/backend@0.1.0 test
> vitest run --run
```

```
RUN v4.1.6 C:/Code/Software Engineering Capstone D424/d424-software-engineering-capstone/backend
```

```
> src/controllers/authController.test.ts [queued]
> src/controllers/dashboardController.test.ts [queued]
> src/controllers/forecastController.test.ts [queued]
> src/controllers/leaseController.test.ts [queued]
> src/controllers/propertyController.test.ts [queued]
> src/controllers/reportsController.test.ts [queued]
> src/controllers/searchController.test.ts [queued]
> src/controllers/tenantController.test.ts [queued]
> src/controllers/transactionController.test.ts [queued]
> src/controllers/unitController.test.ts [queued]
> src/domain/Lease.test.ts [queued]
> src/domain/MaintenanceTicket.test.ts [queued]
> src/domain/mappers.test.ts [queued]
> src/domain/Property.test.ts [queued]
> src/domain/Tenant.test.ts [queued]
> src/domain/Transaction.test.ts [queued]
> src/domain/Unit.test.ts [queued]
✓ src/repositories/UserRepository.test.ts (5 tests) 4ms

> src/controllers/authController.test.ts [queued]
> src/controllers/dashboardController.test.ts [queued]
> src/controllers/forecastController.test.ts [queued]
> src/controllers/leaseController.test.ts [queued]
> src/controllers/propertyController.test.ts [queued]
> src/controllers/reportsController.test.ts [queued]
> src/controllers/searchController.test.ts [queued]
> src/controllers/tenantController.test.ts [queued]
> src/controllers/transactionController.test.ts [queued]
> src/controllers/unitController.test.ts [queued]
> src/domain/Lease.test.ts [queued]
> src/domain/MaintenanceTicket.test.ts 0/9
> src/domain/mappers.test.ts [queued]
> src/domain/Property.test.ts [queued]
> src/domain/Tenant.test.ts [queued]
> src/domain/Transaction.test.ts [queued]
✓ src/domain/Transaction.test.ts (9 tests) 5ms
✓ src/domain/MaintenanceTicket.test.ts (9 tests) 7ms
✓ src/reports/PnLReport.test.ts (4 tests) 7ms
✓ src/middleware/auth.test.ts (7 tests) 5ms
✓ src/reports/MaintenanceAgingReport.test.ts (3 tests) 4ms
✓ src/reports/OccupancyReport.test.ts (3 tests) 13ms
✓ src/forecast/CashFlowForecaster.test.ts (8 tests) 10ms
✓ src/domain/Lease.test.ts (11 tests) 26ms
✓ src/domain/Unit.test.ts (10 tests) 26ms
✓ src/domain/Property.test.ts (13 tests) 25ms
✓ src/domain/Tenant.test.ts (8 tests) 5ms
✓ src/domain/mappers.test.ts (10 tests) 6ms
✓ src/reports/RentRollReport.test.ts (4 tests) 13ms
✓ src/controllers/dashboardController.test.ts (1 test) 4ms
✓ src/repositories/LeaseRepository.test.ts (5 tests) 5ms
✓ src/repositories/MaintenanceTicketRepository.test.ts (6 tests) 6ms
✓ src/repositories/TransactionRepository.test.ts (9 tests) 7ms
✓ src/repositories/UnitRepository.test.ts (7 tests) 6ms
✓ src/repositories/TenantRepository.test.ts (7 tests) 8ms
✓ src/repositories/PropertyRepository.test.ts (12 tests) 8ms
```

```
✓ src/controllers/reportsController.test.ts (5 tests) 6ms
✓ src/controllers/tenantController.test.ts (7 tests) 8ms
✓ src/controllers/transactionController.test.ts (8 tests) 9ms
✓ src/controllers/authController.test.ts (8 tests) 9ms
✓ src/controllers/unitController.test.ts (8 tests) 10ms
✓ src/controllers/leaseController.test.ts (8 tests) 11ms
✓ src/controllers/propertyController.test.ts (8 tests) 12ms
✓ src/controllers/forecastController.test.ts (4 tests) 6ms
✓ src/controllers/searchController.test.ts (2 tests) 8ms
```

```
Test Files 30 passed (30)
  Tests 209 passed (209)
Start at 10:28:13
Duration 2.38s (transform 8.12s, setup 0ms, import 20.02s, tests 281ms, environment 3ms)
```

```
❖ PS C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone> █
```

2.2 Verbose per-test listing

The verbose reporter (`--reporter=verbose`) prints one line per individual case. This is the evidence that each of the 209 cases catalogued in [test-plan.md](#) ran and passed.

```
PS C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone> $OutputEncoding = [System.Text.Encoding]::UTF8
>> [Console]::OutputEncoding = [System.Text.Encoding]::UTF8
>> chcp 65001 | Out-Null
>> npm run test -- -w backend -- --run --reporter=verbose 2>&1 | Select-Object -First 60

> @propertyilot/backend@0.1.0 test
> vitest run --run --reporter=verbose

RUN v4.1.6 C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone/backend

✓ src/domain/Property.test.ts > Property.validate > passes for a fully populated property 2ms
✓ src/domain/Property.test.ts > Property.validate > passes when addressLine2 is null 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when name is blank 1ms
✓ src/domain/Property.test.ts > Property.validate > fails when name exceeds 120 characters 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when addressLine1 is blank 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when city is blank 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when state is lowercase 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when state is too long 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when postalCode is malformed 0ms
✓ src/domain/Property.test.ts > Property.validate > accepts ZIP+4 format 0ms
✓ src/domain/Property.test.ts > Property.validate > aggregates multiple errors at once 0ms
✓ src/domain/Property.test.ts > Property mutators > rename() updates name and bumps updatedAt 14ms
✓ src/domain/Property.test.ts > Property mutators > fullAddress() combines the parts with optional line 2 0ms
✓ src/domain/Unit.test.ts > Unit.validate > passes for a typical unit 3ms
✓ src/domain/Unit.test.ts > Unit.validate > passes with null squareFeet 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when label is blank 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when bedrooms is negative 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when bedrooms is fractional 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when bathrooms is negative 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when marketRent is negative 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when squareFeet is negative 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when propertyId is empty 0ms
✓ src/domain/Unit.test.ts > Unit mutators > setMarketRent updates value and bumps updatedAt 14ms
✓ src/domain/Lease.test.ts > Lease.validate > passes for a typical lease 3ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when startDate equals endDate 0ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when startDate is after endDate 0ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when monthlyRent is zero 0ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when securityDeposit is negative 0ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when unitId or tenantId is blank 0ms
✓ src/domain/Lease.test.ts > Lease behavior > isActiveOn returns true within the lease window when status is ACTIVE 0ms
✓ src/domain/Lease.test.ts > Lease behavior > isActiveOn returns false outside the lease window 0ms
✓ src/domain/Lease.test.ts > Lease behavior > isActiveOn returns false when status is PENDING 0ms
✓ src/domain/Lease.test.ts > Lease behavior > termInMonths approximates the length 0ms
✓ src/domain/Lease.test.ts > Lease behavior > terminate() flips status and bumps updatedAt 14ms
✓ src/reports/OccupancyReport.test.ts > OccupancyReport.generate > counts occupied and vacant units per property and appends a Total row 8ms
✓ src/reports/OccupancyReport.test.ts > OccupancyReport.generate > filters to a single property when propertyId is given 0ms
✓ src/reports/OccupancyReport.test.ts > OccupancyReport.generate > shows 0% occupancy when no leases overlap the asOf date 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > passes for a typical tenant 3ms
✓ src/domain/Tenant.test.ts > Tenant.validate > passes with null phone 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > fails when firstName is blank 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > fails when lastName is blank 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > fails when email is invalid 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > fails when phone has too few digits 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > accepts phone with formatting characters 0ms
✓ src/domain/Tenant.test.ts > Tenant.fullName > joins first and last with a space 0ms
✓ src/reports/RentRollReport.test.ts > RentRollReport.generate > produces one row per lease with property, unit, and tenant joined in 9ms
✓ src/reports/RentRollReport.test.ts > RentRollReport.generate > filters to a single property when propertyId is set 0ms
✓ src/reports/RentRollReport.test.ts > RentRollReport.generate > respects an asOf date by excluding leases not covering that date 0ms
✓ src/reports/RentRollReport.test.ts > RentRollReport.generate > sets title, generatedAt, and toJSON shape 0ms
✓ src/domain/Transaction.test.ts > Transaction.validate > passes for a typical expense 2ms
✓ src/domain/Transaction.test.ts > Transaction.validate > fails when amount is zero 0ms
✓ src/domain/Transaction.test.ts > Transaction.validate > fails when description is blank 0ms
✓ src/domain/Transaction.test.ts > Transaction.validate > fails when propertyId is blank 0ms
PS C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone>
```

3. Results by Layer

The 209 tests break down as follows. The counts come directly from the verbose run shown above.

Layer	Test files	Tests	Notes
Domain classes	6	60	Property, Unit, Tenant, Lease, Transaction, MaintenanceTicket validation + behavior
Mappers	1	10	Round-trip Prisma ↔ domain

Repositories	7	51	CRUD + owner-scoping + search + pagination
Controllers	10	59	Success paths + 404 / ownership / Zod-validation paths
Auth middleware	1	7	Token verification, header parsing, error masking
Reports	4	14	Rent Roll, P&L, Occupancy, Maintenance Aging
Forecast	1	8	12-month cash flow projection
Totals	30	209	Matches the run summary.

4. Results by Test File

Counts taken from the verbose run. Every file shows `pass` .

File	Tests	Status
src/domain/Property.test.ts	13	✓
src/domain/Unit.test.ts	10	✓
src/domain/Tenant.test.ts	8	✓
src/domain/Lease.test.ts	11	✓
src/domain/Transaction.test.ts	9	✓
src/domain/MaintenanceTicket.test.ts	9	✓
src/domain/mappers.test.ts	10	✓
src/repositories/PropertyRepository.test.ts	12	✓
src/repositories/UnitRepository.test.ts	7	✓
src/repositories/TenantRepository.test.ts	7	✓
src/repositories/LeaseRepository.test.ts	5	✓
src/repositories/TransactionRepository.test.ts	9	✓
src/repositories/MaintenanceTicketRepository.test.ts	6	✓
src/repositories/UserRepository.test.ts	5	✓
src/middleware/auth.test.ts	7	✓
src/controllers/propertyController.test.ts	8	✓
src/controllers/unitController.test.ts	8	✓
src/controllers/tenantController.test.ts	7	✓
src/controllers/leaseController.test.ts	8	✓

src/controllers/transactionController.test.ts	8	✓
src/controllers/authController.test.ts	8	✓
src/controllers/dashboardController.test.ts	1	✓
src/controllers/searchController.test.ts	2	✓
src/controllers/reportsController.test.ts	5	✓
src/controllers/forecastController.test.ts	4	✓
src/reports/RentRollReport.test.ts	4	✓
src/reports/PnLReport.test.ts	4	✓
src/reports/OccupancyReport.test.ts	3	✓
src/reports/MaintenanceAgingReport.test.ts	3	✓
src/forecast/CashFlowForecaster.test.ts	8	✓
Total	209	All passing

5. Failures and Resolutions

No tests are currently failing. During development, a small number of authoring mistakes surfaced when the suite was first executed; each was resolved before the final commit. They are listed here for traceability.

#	When	Symptom	Root cause	Fix
1	First reports-controller run	4 assertions failed: expected undefined to be 'Year-to-date Profit & Loss' and similar for the other 3 reports	The test asserted <code>body.name</code> , but <code>Report.toJSON()</code> returns <code>{ title, generatedAt, columns, rows }</code> — the field is named <code>title</code> , not <code>name</code>	Updated <code>reportsController.test.ts</code> to assert <code>body.title</code> for all 4 report controllers

After the fix above was applied, the suite ran clean and has stayed green through subsequent edits.

6. Reproducing These Results

From the repository root on a clean checkout (Node 20+, no database needed):

```
npm install
npm run test -w backend
```

For the per-test listing used in the verbose screenshot:

```
npm run test -w backend -- --run --reporter=verbose
```

The suite has no network or filesystem dependencies and is intended to remain runnable indefinitely as long as Node 20+ and the project dependencies are installed.

7. Summary of Changes Resulting from Testing

Rubric mapping: Task 3.D.4.

No production bugs were uncovered by the new test suite — the application code was already correct. Writing the tests did, however, surface three small refactorings that improved the codebase. Each is described below with the motivation and the resulting change.

7.1 Tightened the Authorization header parsing contract

Trigger. Writing `auth.test.ts` made it obvious how many distinct failure shapes a malformed header has — missing entirely, wrong scheme, right scheme but empty token, and a verifier that throws an internal error containing sensitive text.

Change. The middleware's existing branches were left in place, but a new test (`TC-MW-AUTH-07`) was added asserting that the message handed to `next()` does not contain the raw verifier exception. The middleware was already masking the underlying error in the catch block (`next(new HttpError(401, 'Invalid or expired token'))`), but the masking was an implicit choice. The test now pins that behavior so a well-intentioned refactor that forwards the original error message cannot regress the contract by accident.

File touched: `backend/src/middleware/auth.test.ts` (new test added; middleware unchanged).

7.2 Made the ownership-scoping contract explicit and uniform

Trigger. When the repository tests were drafted file-by-file, the scoping patterns turned out to be subtly different across entities: `PropertyRepository` and `TenantRepository` scope directly by `ownerId`; `UnitRepository`, `TransactionRepository`, and `MaintenanceTicketRepository` scope through `property.ownerId`; and `LeaseRepository` scopes through `unit.property.ownerId` (two-level join).

Change. Rather than refactoring the production code (each pattern is correct for its model), the test suite was structured so every repository has a `findById` test whose only purpose is to verify the exact `where` clause that gets sent to Prisma. That puts the scoping shape under version control as a contract test, so an accidental drop of the join during a future refactor would fail immediately and visibly.

Files touched: all 7 repository test files contain a scoping case near the top of the file.

7.3 Pinned default page sizes and validation defaults so they cannot drift silently

Trigger. While writing `PropertyRepository.test.ts`, I noticed that the default `pageSize` (20) and the `propertyType` default for `Property.create` were only documented in code comments. A future change that flipped the default to 50 or to a different enum value would not have failed any test.

Change. Added explicit assertions for both defaults — `PropertyRepository.list` returns `pageSize: 20` when the caller omits it (`TC-RP-PROP-03`), and the `Property.create` factory paths in the controller tests assume the existing `'SINGLE_FAMILY'` default. These assertions are cheap to maintain and force any future change to those defaults to be deliberate.

Files touched: `backend/src/repositories/PropertyRepository.test.ts`,
`backend/src/controllers/propertyController.test.ts`.

8. Sign-off

Item	Status
All planned test cases authored	✓
Suite runs green on a clean checkout	✓
Screenshots captured and committed	✓
Summary of changes documented	✓ (section 7 above)